

Aussie Primary Academy | Australian Curriculum v9.0 | Learn, Investigate and Explain

Name: _____

Date: _____

LEARNING INTENTION	consider how people use scientific explanations to meet a need or solve a problem
KEY IDEA	A science-based solution connects an explanation to a need, then tests effectiveness, safety and limitations. The strongest choice may involve trade-offs such as cost, access, durability and environmental impact.
WORKED MODEL	Knowing that some materials conduct heat slowly helps people choose insulation. Tests then compare temperature change, safety, cost and reuse before a material is recommended.

SUCCESS CRITERIA - I CAN

☐ identify need and science idea ☐ compare solutions ☐ use criteria and evidence ☐ justify a recommendation

A. BUILD THE SCIENCE IDEA

1. Match need, scientific explanation, solution and test evidence in an insulation example.

2. Explain how understanding material properties can guide a design.

3. List two reasons a solution that works in a test may not suit every community.

B. APPLY AND EXPLAIN

4. Compare three lunch-box linings using heat loss, strength, cost and waste.

5. Select the most relevant evidence for the need and reject one irrelevant preference.

6. Write a recommendation that includes one benefit and one limitation.

C. INDEPENDENT CHECK

7. Construct a decision table with criteria, evidence and rating for three options.

8. Check whether the table considers safety, access and environmental impact.

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D. APPLY SCIENCE IN CONTEXT

9. A footpath becomes dangerously hot. Tests show pale coating lowers surface temperature 5 C, shade lowers it 9 C and watering lowers it 7 C briefly.

10. Recommend one option or combination for a dry community and justify with science plus two constraints.

11. State the monitoring data needed to decide whether the response works over time.

E. REASON, EVALUATE AND TRANSFER

12. Evaluate: The solution with the largest immediate effect is automatically best.

13. Design a small test of another everyday solution and name the evidence that would count as success.

F. STUDENT REVIEW - SHOW WHAT YOU UNDERSTAND

[] I can connect a scientific explanation to an evidence-based solution.

Circle one: independently | with one reminder | with support

Evidence from my work that proves I understand: _____

The most difficult science idea was _____ because _____

One correction or improvement I made: _____

A question I still have: _____

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Accept equivalent responses when the student uses accurate science ideas, relevant evidence and clear reasoning.

COMPLETE ANSWER KEY

1. Need=keep contents cool; explanation=slow heat transfer; solution=insulating lining; evidence=temperature change.
2. Properties predict performance and guide fair tests.
3. Contexts differ in cost, access, climate, safety, maintenance or values.
- 4-6. Recommendation must use comparable data and acknowledge a limitation.
7. Criteria and evidence should be applied consistently to all options.
8. Add any missing stakeholder or consequence criterion.
9. Shade has the largest sustained stated reduction; watering is temporary and uses water.
10. Shade or shade plus coating is defensible; reasoning must address dry conditions and constraints.
11. Repeated surface temperature, coverage, durability, maintenance, cost and user safety.
12. False; duration, risk, feasibility and consequences also matter.
13. Test must define need, changed factor, measurable success and safe method.

TEACHER CHECK AND NEXT STEP

[] Uses accurate vocabulary and a complete cause-and-effect explanation.

[] Supports conclusions with observations or data.

Next step: revisit one misconception, then transfer the idea to a new example.

COMMON MISCONCEPTION / WATCH OUT

Science informs a decision but does not erase practical and community considerations.

A solution should be evaluated against the stated need, not personal preference.

EXTENSION AND RE-TEACH

Re-teach: model one difficult item, complete one together, then ask the student to explain an independent correction.

Extension: apply the science idea to a new real-world example and justify the explanation with observations or data.